

# MATH 350-2 Advanced Calculus

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# Outline

- 1 Real Analysis Lecture 8
  - Compactness
  - Metric Spaces

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# Warm-up Challenge

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- Lindelöf Covering Theorem

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about real numbers.

## Theorem (Bolzano-Weierstrass Theorem)

*A bounded set  $A \subseteq \mathbb{R}^n$  with infinitely many points must have an accumulation point.*

# Cantor Intersection Theorem

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*Suppose that  $C_1, C_2, C_3, \dots$  are non-empty closed, bounded sets with  $C_{i+1} \subseteq C_i$  for all  $i \in I = \mathbb{Z}_+$ . Then  $\bigcap_{i \in I} C_i$  is also non-empty.*

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Therefore the Bolzano-Weierstrass Theorem tells us it has an accumulation point  $\vec{x}$ .





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$$C_m = A \cap \left( \mathbb{R}^n \setminus \bigcup_{i=1}^m U_i \right).$$

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This is a contradiction.



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The value  $d(x, y)$  is called a **metric** and describes the “distance” between  $x$  and  $y$ .



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## Definition

Let  $(M, d)$  be a metric space. The open ball of radius  $r > 0$  centered at  $x \in M$  is

$$B_M(x; r) = \{y \in M : d(x, y) < r\}.$$

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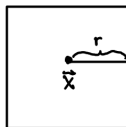
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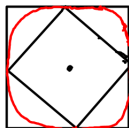
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# Challenge!

Let  $M$  be a nonempty set and define  $d : M \times M \rightarrow \mathbb{R}$  by

$$d_{\text{disc}}(x, y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$$

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This is called the **discrete metric**.

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What do the open balls with the discrete metric look like?

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The set  $A$  is **open** if every point in  $A$  is an interior point, or equivalently  $\text{int}(A) = A$ .

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## Problem

Consider the metric space  $(\mathbb{R}, d_{\text{disc}})$  where  $d_{\text{disc}}$  is the discrete metric.

What sets are open sets?

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The set  $\bar{A}$  of all adherent points of  $A$  is called the **closure** of  $A$ .

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Prove the following (Apostol Theorem 3.36). If  $(M, d)$  is a metric space,  $U \subseteq M$  is open, and  $C \subseteq M$  is closed, then  $U \setminus C$  is open and  $C \setminus U$  is closed.

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- (d)  *$A = \bar{A}$*

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